1

One *can* burn a diamond in air.

2

The “strength” is a dimensionless measure of the coupling constant involved in each interaction (∼ means “of the order”).

3

How I’m rushing through this! How much each sentence in this brief story contains. “The stars are made of the same atoms as the earth.” I usually pick one small topic like this to give a lecture on. Poets say science takes away from the beauty of the stars—mere globs of gas atoms. Nothing is “mere.” I too can see the stars on a desert night, and feel them. But do I see less or more? The vastness of the heavens stretches my imagination—stuck on this carousel my little eye can catch one-million-year-old light. A vast pattern—of which I am a part—perhaps my stuff was belched from some forgotten star, as one is belching there. Or see them with the greater eye of Palomar, rushing all apart from some common starting point when they were perhaps all together. What is the pattern, or the meaning, or the *why?* It does not do harm to the mystery to know a little about it. For far more marvelous is the truth than any artists of the past imagined! Why do the poets of the present not speak of it? What men are poets who can speak of Jupiter if he were like a man, but if he is an immense spinning sphere of methane and ammonia must be silent?

4

Our point here is not so much the result, (4.3), which in fact you may already know, as the possibility of arriving at it by theoretical reasoning.

5

Counting antibaryons as—1 baryon.

6

A radius vector is a line drawn from the sun to any point in a planet’s orbit.

7

That is, how far the circle of the moon’s orbit falls below the straight line tangent to it at the point where the moon was one second before.

8

The proof is not given in this course.